Assignment No: 2

Q1. Hollow Square

* (1,1)	* (1,2)	* (1,3)	* (1,4)	* (1,5)
* (2,1)	o (2,2)	o (2,3)	o (2,4)	* (2,5)
* (3,1)	o (3,2)	o (3,3)	o (3,4)	* (3,5)
* (4,1)	o (4,2)	o (4,3)	o (4,4)	* (4,5)
* (5,1)	* (5,2)	* (5,3)	* (5,4)	* (5,5)

n = 5

Dry Run

```
for( int i = 1 ; i<=n ; i+1){
    for(int j = 1 ; i<=n ; j+1){
        if( 2<=j<=4 && 2<=j<=4){
            Soprint(" ");
        }else{
            Soprint("* _ ");
        }
    }
}
```

i<=5
→ j<=5
→ *
2<=5
→ *
3<=5
→ *
4<=5
→ *
5<=5
→ *

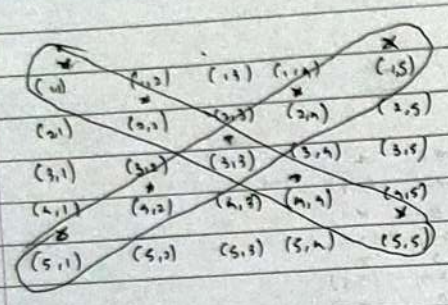
2<=5
→ j<=5
→ *
2<=5
→ o
3<=5
→ o
4<=5
→ o
5<=5
→ *

Code.

```
public static void main(String[]args){
    Scanner sc = new Scanner(System.in);
    int n = sc.nextInt();
    for (int i = 1; i<=n; i++){
        for(int j = 1; j<=n; j++){

            if(2<=i<=4 && 2<=j<=4){
                Soprint(" ");
            }else{
                soprint("* _ ");
            }
        } soprintln(" ");
```

2.



•Dry run n=5

i=1 , i<=n , i++

1

1<=5

→ 1<=5

→ (i==j || i+j=6)

→ *

2<=5

→ 0

3<=5

→ 0

4<=5

→ 0

5<=5 → * (i==j || i+j=~~7~~6)

```
public static void main(String[]args){
    Scanner sc = new Scanner(System.in);
    int n = sc.nextInt();

    for(int i=1 ; i<=n ; i++){
        for(int j=1 ; j<=n ; j++){

            if(i==j || i+j==n+1){
                System.out.print("* ");
            }else{
                System.out.print(" ");
            }
        }
    }
}
```

Q3

```
*
* *
* * *
* * * *
* * * * *
```

•Dry run n = 5 ; i<=n ; i++

1<=5

→ i>=1

→ i>=j

→ 1>=1

→ *

2>=2

→ 0

2>=1

→ 2>=1

→ *

2>=2

→ *

2>=3

→ 0

```
public static void main(String[]args){
    Scanner sc = new Scanner(System.in);
    int n = sc.nextInt();
    for(int i=1 ; i<=n ; i++){
        for(int j=1 ; j>=i ; j++){

            System.out.print("* ");
        }
        System.out.println(" ");
    }
}
```

Q4

```
* * * * *
* * * *
* * *
* *
*
```

*1) *Dry run n=5 i=1; i<=5; i++

i<=5
↳ 5>=1, j--
↳ *

4>=1
↳ *

3>=1
↳ *

2>=1
↳ *

1>=1
↳ *

2<=5

5>=2
↳ *

4>=2
↳ *

3>=2
↳ *

2>=2
↳ *

```
public static void main(String[] args){
    for(int i=1; i<=5; i++){
        for(int j=n; j>=i; j--){
            System.out.print("* ");
        }
        System.out.println(" ");
    }
}
```

* n-i+1
5-1+1 = 5
5-2+1 = 4

Q5.

```
* # * #
# * # *
* # * #
# * # *
```

*Dry run i=1; i<=n; i++

1<=4
↳ 1<=4, j++
↳ if((i+j)%2==0){ System.out(*) }
↳ *

2<=4
↳ #

```
for(int i=1; i<=n; i++){
    for(int j=1; j<=n; j++){
        if((i+j)%2==0){
            System.out.print("*");
        }else{
            System.out.print("#");
        }
    }
    System.out.println("#");
}
```

ex.6

```
    *
   * *
  * * *
 * * * *
* * * * *
```

n=5

Dry run

```
for(int i=1 ; i<=5; i++){
    for(int j=1 ; j<=5; j++){
        if(j>=n-i+1){
            printf("*");
        }else{
            printf(" ");
        }
    }
}
```

i<=5
↳ ~~j<=5~~
↳ j<=n-i+1
~~i<=4~~
printf(*)

2<=5
↳ 4=5-2+1

i<=5
↳ i<=5
↳ j==5-1+1
1

- - - *
* *

ex 7

```
* * * * *
 * * * *
  * * *
   * *
    *
```

i=1
↳ j<=5
↳ i>=j
↳

```
if(i>j){
    print(-)
}else{
    (*)
}
```